IDEAS · RISK · PHILOSOPHY

Is Trusting Nature a Fallacy? Taleb, Kahneman & Tversky, and Popper

A page from Antifragile makes a bold claim: when you're unsure about risk, the bigger mistake is dismissing nature. Two other traditions of thought help show whether that claim holds up.

Reading notes on Nassim Nicholas Taleb, *Antifragile*, "Modernity and the Denial of Antifragility," p. 124

Few phrases end an argument faster than "that's the naturalistic fallacy." Mention that humans have eaten a food for thousands of years, or that a remedy is old, and someone will remind you that "natural" doesn't mean "good." They're right, up to a point. On one dense page of *Antifragile*, Nassim Taleb argues that the reply has itself become a mistake, and a more dangerous one.

I want to unpack his argument, then test it against two of the most influential frameworks for thinking about thinking: the heuristics-and-biases research of **Daniel Kahneman and Amos Tversky**, and the falsificationism of **Karl Popper**.

01 What Taleb actually says

Taleb starts by agreeing with the philosophers. The *naturalistic fallacy* is the error of assuming that what is natural is therefore morally right. Nature is full of things we shouldn't copy, and he says plainly that he subscribes to this view.

He also criticizes one familiar misuse of nature: applying Darwinian selection to society and concluding that we shouldn't protect those who fail. That moves from "this is how nature works" to "this is how we ought to behave," which is exactly the jump the naturalistic fallacy warns against.

Then he turns the argument around. The problem, he writes, is that some people take the fallacy *outside the moral domain* and use it to dismiss any reliance on nature, even when we're in doubt. On questions of risk, nature isn't a fallacy at all:

> *"Time is the best test of fragility ... and nature is the only system that has been stamped 'robust' by time."*
>
> — Taleb, *Antifragile*, p. 124

He goes further. The "philosophasters," his word for pseudo-philosophers, fail to understand that risk and survival come before philosophizing, and they "should eventually exit the gene pool." His conclusion:

> *“There is a worse fallacy: people making the opposite mistake and considering that what is natural is a fallacy.”*

IN ONE SENTENCE

Nature doesn't tell you what is morally right, but it is a strong guide to what is *safe*. Treating its long survival record as a “fallacy” is the more dangerous mistake.

02 The Kahneman & Tversky challenge

Kahneman and Tversky built a research program showing that human judgment relies on mental shortcuts, or *heuristics*, that go wrong in predictable ways. Seen through their work, Taleb's argument looks vulnerable at first.

“Natural = safe” may just be a shortcut

Preferring the natural can be exactly the kind of gut rule their work warns about. It *feels* safe, so we judge it safe. But arsenic, hemlock and earthquakes are natural too.

Status quo bias

People tend to prefer what already exists and treat any change as a loss. “Stick with what's old” could be that bias presented as wisdom.

Survivorship bias

“Stamped robust by time” counts only the survivors. More than 99% of the species that ever lived are extinct. Nature as a *system* has been robust, but that doesn't mean any single natural thing is.

03 ...and where their work supports Taleb

On a closer look, much of Kahneman and Tversky's research supports Taleb more than it undermines him.

The law of small numbers

Just before this passage, Taleb warns about the consequences of “jumping on information.” That matches Tversky and Kahneman's classic finding, the *belief in the law of small numbers*: people, including trained scientists, give far too much weight to small, new samples. A new nutrition study is a small sample. Thousands of years of human experience is an enormous one. The bias is overreacting to the study, not trusting the long record.

Stories versus base rates

A clever theory about why something new is perfectly safe can persuade because it's a good story, not because the evidence is strong. The “philosophasters” are building convincing arguments while ignoring the base rate, which here is the track record.

Loss aversion and ruin

This is the most interesting point of conflict. Prospect theory found that people feel losses more strongly than equal gains, and this loss aversion is usually listed as a bias. Taleb disagrees. When a loss could mean *ruin* (death, bankruptcy, extinction), extreme caution is rational, because once you're ruined you can't collect any future gains. That is what he means by putting survival before philosophizing.

Some “biases” may be sensible behaviour in a world of fat tails and losses you can't undo.

This view is closer to Gerd Gigerenzer's idea of *ecological rationality*: heuristics that have lasted usually lasted for a reason.

04 The Popperian backbone

If Kahneman and Tversky supply the objections, Karl Popper supplies much of the foundation. Taleb borrows heavily from him, and this passage shows it.

Nature as a theory that keeps surviving tests

For Popper, a theory is never proven true. It can only survive serious attempts to refute it, and each test it survives *corroborates* it, provisionally. Seen this way, nature is a huge collection of conjectures (organisms, behaviours, ecosystems) that has been tested against reality for billions of years. No human theory has faced that many tests. When Taleb calls time “the best test of fragility,” he is applying Popper's idea to survival.

Letting ideas die instead of us

Popper said that the difference between the amoeba and Einstein is that the amoeba dies with its mistaken guesses, while the scientist lets the *theories* die in their place. That's why we test ideas in laboratories rather than on our own lives. Taleb's remark that philosophasters will “exit the gene pool” reverses this: if you ignore risk, you aren't letting your theory die for you. You are the one at risk.

The asymmetry of evidence

A million white swans can't prove that all swans are white, but one black swan disproves it. Taleb's earlier book *The Black Swan* is built on this asymmetry. Applied here, “we've seen no evidence of harm” from a new chemical is weak, because it has faced few tests. “Humans have eaten this for ten thousand

years" is strong, because it has faced many. **The burden of proof falls on the new thing, not on nature.**

Popper agrees on morality too

In *The Open Society and Its Enemies*, Popper defended a strict separation between facts and decisions: you cannot derive moral norms from how the world is. He also opposed social Darwinism. So he would back Taleb on both points. Nature isn't a guide to what is right, but it is a strong guide to what survives.

Piecemeal tinkering

Popper preferred small, reversible "piecemeal" reforms to grand utopian plans, because small errors can be found and fixed. This is close to Taleb's preference for trial and error over top-down design.

05 Where both traditions push back

Combine Popper with Kahneman and Tversky and you get the strongest objection to Taleb:

> THE MISMATCH PROBLEM
>
> A theory is only corroborated for the tests it actually faced.

Nature was tested in *ancestral* environments, not modern ones. Our craving for sugar helped when food was scarce and harms us now that it's everywhere. Biologists call this *evolutionary mismatch*. So "natural" is well tested in its original context and less trustworthy once that context changes.

Taleb mostly anticipates this, which is why he limits his claim to cases where we are *in doubt*. Nature is the default, the starting position that new things must improve on. It isn't beyond question.

06 The three views side by side

	Kahneman & Tversky	Popper
Core idea	Heuristics produce systematic errors	Knowledge grows by surviving attempts at refutation
Challenge to Taleb	"Natural = safe" may be status quo or survivorship bias	Corroboration only holds in the environment where the tests happened
Support for Taleb	Overreacting to small, new data is the real bias	Nature is the most-tested "theory" we have
Key tension	Loss aversion as a bias vs. ruin-avoidance as rational	Letting theories die instead of us vs. people "exiting the gene pool"
On morality	Largely silent	Agrees: norms can't be derived from facts

07 So, is trusting nature a fallacy?

It depends on the question. If you're asking what is *right*, then yes: nature is no moral authority, and Taleb agrees. If you're asking what is *safe* and you're unsure, then usually no. Kahneman and Tversky remind us that trusting the familiar can be a bias. Popper explains why, under real uncertainty, relying on what has survived the most tests is often the most rational choice available.

The practical lesson is simple. Next time someone answers “it's been done this way for millennia” with “that's the naturalistic fallacy,” ask one question: *are we talking about ethics, or about risk?* The answer decides which side has the burden of proof.

Sources discussed: N. N. Taleb, *Antifragile* (2012) and *The Black Swan* (2007); A. Tversky & D. Kahneman, “Belief in the Law of Small Numbers” (1971) and “Prospect Theory” (1979); D. Kahneman, *Thinking, Fast and Slow* (2011); K. Popper, *The Open Society and Its Enemies* (1945) and *Objective Knowledge* (1972); G. Gigerenzer on ecological rationality.